

Gregory Ryan Seaton

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Summary

Machine learning and AI researcher focused on optimizing workflows in modern ML pipelines. Experienced in model development and designing efficient, scalable processes that improve deployment outcomes, with an emphasis data-driven decision optimization.

Education

Washington University in St Louis: Bachelor of Science, Data Science

August 2023 - May 2027

- **GPA:** 3.88
- **Achievements:** Danforth Scholar, Synchrony Banking Scholar, Bauer Leadership Program Inductee, Delta Sigma Pi Brother
- **Coursework:** Data Science Statistics, Data Structures & Algorithms, Linear Algebra, Decision and Estimation Theory for Stochastic Processes, Optimization, Linear Statistical Models, Complex Network Analysis, GIS for Spatial Science, Artificial Intelligence
- **Skills:** Python, TensorFlow, PyTorch, SQL, R, Excel, MongoDB, Tableau, ArcGIS, Bash, Java, React, HTML, MATLAB, GitHub, Dynamic & Integer Programming, Convex/Linear/Bayesian Optimization, Stochastic Dynamical Systems

Work Experience

Electric Power Engineers

May 2026 – Present

Software Power Systems Engineering Intern (Data Science Track)

Austin, Texas

- Architected an end-to-end power-grid entity-resolution and clustering pipeline across 490K+ PSS/E network and 101K+ geospatial records, leveraging Union-Find, DBSCAN, and Bayesian hyperparameter optimization to reconcile records and improve ARI by 0.40.
- Automated GIS detection and reconciliation of newly introduced power-grid buses, reducing a 2-hour manual discovery workflow to near-real-time detection and generating \$40K in annual productivity savings through streamlined network integration.
- Prototyped an LLM-based expert recommendation system for a 380+ engineering organization, using TF-IDF and semantic similarity to classify project technical areas and rank subject-matter experts, with an estimated \$50K in potential annual productivity savings.

National Science Foundation: IIT Madras & ePlane Company

May 2025 – August 2025

International ML Research Fellow

Chennai, Tamil Nadu, India

- Designed an ML-driven modeling framework to learn and predict operational patterns in autonomous air-taxi networks, using data-centric pipelines to capture dispatch behavior, battery usage, and vertiport capacity constraints across Indian urban regions.
- Built a stochastic demand-modeling pipeline using Poisson processes and temporal feature engineering to simulate 400,000 daily trips.
- Developed a TensorFlow-based geospatial clustering model on taxi GPS data to identify 11 high-demand hubs and translated cluster outputs into downstream-compatible ML features for deployment planning and user-centric business experimentation.
- Delivered a 30%+ decrease in average wait times through a decision-optimization engine that used a global scoring model to evaluate feasible aircraft-request pairings, improve assignment quality, and maintaining system-wide fleet availability.

University of Utah

May 2024 – May 2025

Systems Engineering Research Intern

Salt Lake City, Utah

- Formalized an AWS autonomous-vehicle security testbed that increased experimental reproducibility by 45% (measured by reduction in run-to-runs variance) and improved the reliability of cyber-physical system evaluations.
- Designed and conducted 100+ adversarial ML experiments, demonstrating computer vision classification-confidence drops of 40–60% under physical-world perturbations, and identifying critical perception vulnerabilities.

Beats By Dre

Jul 2024 - Sep 2024

Consumer Insight Analyst Extern

Remote

- Conducted NLP-driven sentiment analysis across 1,000 Amazon reviews spanning 10 audio product lines, building an automated web-scraping and VADER text-processing pipeline to benchmark consumer sentiment and identify insights for Beats and competing brands.

Projects

Geospatial Vessel Routing Optimization

Apr 2025

- Developed a multi-factor geospatial optimization model in ArcGIS Pro integrating 300K+ AIS vessel records, historical hurricane tracks, and ocean-current vectors into a weighted cost framework to identify least-cost vessel routing across the Gulf of Mexico.

Parkinson's Disease Detection Using Voice-Based Machine Learning Models

Apr 2025

- Implemented TensorFlow CNN to classify Parkinson's disease from biomedical voice features, achieving 94% diagnostic accuracy.

Leadership

Data Science Network

May 2024 - Present

Founder, President

- Founded a 185+ member student organization focused on data science career exploration and interdisciplinary learning

Washington University in St. Louis Admissions

Jan 2025 - Present

Bear Ambassador

- Deliver engaging and informative campus tours to over 400+ prospective students and families

Data Structures and Algorithms Curricula Team

Jul 2024 - Jan 2025

Teaching Assistant

- Conducted weekly sessions with 15 students to practice implementing data structures throughout the academic year
- Evaluated over 520 assignments and exams while offering constructive feedback to help students improve their performance

Interests

International Solo Traveling, Pokémon, Greenery, Soccer, Hiking, Meditation, Reading, Board Games